

THE MONUMENT, Australia

WMO: 943360

Lat: 21.8125S

Lon: 139.9267E

Elev: 948

StdP: 14.20

Time Zone: 10.00 (E10)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>7      | (b)<br>41.3 | (c)<br>43.8 | (d)<br>11.3                   | (e)<br>10.1 | (f)<br>64.5 | (g)<br>13.9 | (h)<br>11.5 | (i)<br>64.7 | (j)<br>19.2           | (k)<br>66.6 | (l)<br>18.0 | (m)<br>66.4 | (n)<br>4.1            | (o)<br>180 | (p)<br>0.469 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 20.3                   | 105.8           | 67.8 | 104.0 | 67.6 | 102.1 | 67.4 | 75.9                | 87.3 | 75.0 | 86.1 | 74.2 | 85.7 | 10.5                 | 90   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 73.3                            | 128.2 | 79.3 | 72.3 | 124.0 | 79.5 | 71.3 | 119.5 | 79.8 | 40.1          | 87.8 | 39.3 | 85.6 | 38.5 | 85.0 | 88.0              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|-------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
| Min               |      | Max  |                            | Min  | Max  | Min                | Max                                                | Min       | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 20.3              | 18.2 | 16.5 |                            | DB   | 35.7 | 109.1              | 1.8                                                | 2.6       | 34.4 | 111.0      | 33.4 | 112.5      | 32.3 | 114.0      | 31.0 | 115.9 |
|                   |      |      |                            | WB   | 30.4 | 78.2               | 1.9                                                | 3.0       | 29.0 | 80.4       | 27.9 | 82.2       | 26.8 | 83.9       | 25.4 | 86.1  |

Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 76.1        | 87.9  | 85.7  | 83.4  | 77.0  | 68.4  | 61.3  | 60.9  | 64.4  | 72.7  | 80.0  | 84.3  | 87.5  | (6)  |
|                                                                        | DBStd       | 11.45       | 5.15  | 5.09  | 4.95  | 5.30  | 6.00  | 6.11  | 5.83  | 6.52  | 7.31  | 6.56  | 5.76  | 5.47  | (7)  |
|                                                                        | HDD50       | 1           | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | (8)  |
|                                                                        | HDD65       | 444         | 0     | 0     | 0     | 2     | 34    | 146   | 155   | 91    | 15    | 1     | 0     | 0     | (9)  |
|                                                                        | CDD50       | 9518        | 1174  | 999   | 1035  | 810   | 569   | 340   | 338   | 448   | 681   | 931   | 1030  | 1163  | (10) |
|                                                                        | CDD65       | 4486        | 709   | 579   | 570   | 362   | 138   | 35    | 28    | 74    | 246   | 467   | 580   | 698   | (11) |
|                                                                        | CDH74       | 62887       | 10446 | 8162  | 7566  | 4329  | 1512  | 399   | 437   | 1171  | 3414  | 6553  | 8462  | 10436 | (12) |
|                                                                        | CDH80       | 36130       | 6501  | 4863  | 4251  | 2103  | 475   | 68    | 72    | 405   | 1717  | 3815  | 5208  | 6652  | (13) |
| (14) Wind                                                              | WSAvg       | 8.8         | 9.6   | 9.4   | 9.1   | 8.3   | 8.0   | 8.1   | 7.8   | 8.4   | 8.9   | 9.4   | 9.4   | 9.5   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 13.1        | 3.0   | 3.0   | 1.1   | 0.4   | 0.6   | 0.4   | 0.3   | 0.2   | 0.4   | 0.4   | 1.1   | 2.0   | (15) |
|                                                                        | PrecMax     | 27.4        | 12.8  | 7.6   | 7.9   | 4.2   | 3.6   | 2.8   | 1.5   | 1.7   | 4.6   | 2.6   | 2.8   | 4.1   | (16) |
|                                                                        | PrecMin     | 3.9         | 0.1   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.3   | (17) |
|                                                                        | PrecStd     | 6.0         | 3.0   | 2.2   | 1.7   | 0.9   | 0.9   | 0.7   | 0.5   | 0.4   | 0.9   | 0.6   | 0.9   | 1.2   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 108.1 | 105.9 | 103.9 | 97.5  | 92.0  | 85.3  | 85.7  | 92.7  | 99.9  | 104.1 | 106.9 | 109.0 | (19) |
|                                                                        |             | MCWB        | 69.0  | 68.3  | 68.4  | 66.2  | 63.9  | 61.0  | 60.5  | 61.5  | 63.8  | 65.5  | 66.5  | 69.5  | (20) |
|                                                                        | 2%          | DB          | 105.1 | 102.9 | 100.5 | 95.0  | 87.7  | 81.8  | 81.7  | 88.5  | 96.3  | 101.4 | 103.9 | 105.8 | (21) |
|                                                                        |             | MCWB        | 69.0  | 68.8  | 68.3  | 65.7  | 62.2  | 60.4  | 58.4  | 59.4  | 62.2  | 64.5  | 66.0  | 67.5  | (22) |
|                                                                        | 5%          | DB          | 102.7 | 100.6 | 98.2  | 92.8  | 85.2  | 79.0  | 79.0  | 84.6  | 93.2  | 98.9  | 101.6 | 103.7 | (23) |
|                                                                        |             | MCWB        | 68.6  | 68.3  | 67.5  | 64.0  | 61.2  | 59.0  | 57.0  | 57.7  | 61.0  | 63.5  | 65.5  | 67.5  | (24) |
|                                                                        | 10%         | DB          | 100.1 | 98.4  | 95.8  | 90.4  | 82.3  | 75.3  | 75.8  | 80.7  | 89.8  | 95.8  | 99.0  | 101.2 | (25) |
|                                                                        |             | MCWB        | 68.8  | 68.1  | 67.1  | 63.3  | 59.5  | 56.8  | 55.7  | 55.6  | 59.5  | 62.6  | 65.1  | 67.3  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 77.2  | 76.4  | 76.3  | 72.8  | 69.2  | 64.9  | 66.3  | 65.2  | 68.3  | 70.3  | 73.1  | 76.1  | (27) |
|                                                                        |             | MCDB        | 89.5  | 89.2  | 85.6  | 85.3  | 82.5  | 78.1  | 77.6  | 83.9  | 83.1  | 88.4  | 90.0  | 89.6  | (28) |
|                                                                        | 2%          | WB          | 76.2  | 75.2  | 74.7  | 70.8  | 65.7  | 63.0  | 62.8  | 62.2  | 65.8  | 68.2  | 71.4  | 74.5  | (29) |
|                                                                        |             | MCDB        | 87.6  | 87.5  | 83.5  | 82.5  | 81.0  | 75.8  | 74.3  | 81.3  | 83.9  | 87.4  | 89.4  | 87.5  | (30) |
|                                                                        | 5%          | WB          | 75.3  | 74.3  | 73.7  | 68.5  | 63.6  | 61.0  | 59.6  | 59.4  | 63.5  | 66.6  | 70.1  | 73.5  | (31) |
|                                                                        |             | MCDB        | 86.2  | 86.6  | 82.8  | 82.3  | 79.7  | 74.0  | 74.2  | 81.7  | 86.2  | 88.4  | 87.4  | 87.1  | (32) |
|                                                                        | 10%         | WB          | 74.5  | 73.5  | 72.5  | 66.6  | 61.7  | 59.0  | 57.0  | 57.2  | 61.4  | 64.9  | 69.0  | 72.6  | (33) |
|                                                                        |             | MCDB        | 85.5  | 85.8  | 83.6  | 84.0  | 78.9  | 72.8  | 73.8  | 77.7  | 86.4  | 90.2  | 87.1  | 87.4  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 20.3  | 21.0  | 21.2  | 23.9  | 24.9  | 24.7  | 26.6  | 28.4  | 28.4  | 27.5  | 25.9  | 23.3  | (35) |
|                                                                        |             | MCDBR       | 24.6  | 24.3  | 24.3  | 25.7  | 26.6  | 27.7  | 30.1  | 31.4  | 30.2  | 29.8  | 28.6  | 26.3  | (36) |
|                                                                        | 5% WB       | MCWBR       | 7.9   | 7.4   | 8.4   | 9.4   | 11.5  | 12.9  | 14.2  | 14.0  | 13.1  | 12.0  | 10.6  | 9.2   | (37) |
|                                                                        |             | MCWBR       | 17.2  | 17.5  | 16.3  | 19.0  | 21.9  | 22.0  | 24.0  | 27.0  | 23.9  | 23.7  | 22.9  | 19.5  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.384 | 0.376 | 0.356 | 0.340 | 0.327 | 0.310 | 0.310 | 0.322 | 0.364 | 0.373 | 0.391 | 0.395 | (40) |
|                                                                        |             | taud        | 2.462 | 2.481 | 2.514 | 2.512 | 2.493 | 2.516 | 2.489 | 2.432 | 2.343 | 2.340 | 2.350 | 2.385 | (41) |
|                                                                        | Ebn at Noon |             | 305   | 303   | 301   | 293   | 284   | 284   | 287   | 294   | 294   | 300   | 300   | 301   | (42) |
|                                                                        |             | Edh at Noon | 38    | 37    | 34    | 32    | 30    | 28    | 29    | 34    | 40    | 42    | 43    | 41    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 2253  | 2260  | 2075  | 1847  | 1528  | 1376  | 1496  | 1806  | 2098  | 2338  | 2382  | 2333  | (44) |
|                                                                        | RadStd      |             | 212   | 148   | 154   | 141   | 91    | 88    | 84    | 73    | 138   | 156   | 119   | 181   | (45) |

Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (5 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |